

Chapter 2: Airworthiness & MEL Systems

Syllabus Source: CAE Oxford (pp. 53-58) | Pass Target: >= 90% AviationExam

1. REGULATORY FRAMEWORK: ANNEX 8 VS ANNEX 6

Airworthiness is the intrinsic safety of an aircraft to operate safely within authorized flight operating limits. Under the ICAO regulatory framework, airworthiness is split across two core Annexes:

- Annex 8 (Airworthiness of Aircraft): Deals with technical, physical, and design standards: structural integrity, propulsion reliability, systems crashworthiness, avionics testing, and continuing airworthiness.
- Annex 6 (Operation of Aircraft): Deals with operational safety standards: flight instruments, operating limitations, emergency equipment, performance operating minimums, and maintenance operational management.

[*] OFFICIAL DEFINITION - Annex 8 Applicability Scope (EASA ECQB Core)

The international airworthiness standards of ICAO Annex 8, Part III apply to aeroplanes with a Maximum Take-Off Mass (MTOM) GREATER THAN 5,700 kg, powered by at least TWO engines, and intended for the international carriage of passengers, cargo, or mail.

2. CERTIFICATION CHAIN: PROTOTYPE TO COMMERCIAL AIRLINE

An aircraft design and individual airframe pass through strict legal certification stages:

Certificate	Issuing Authority	Legal Scope & Operational Purpose
1. Permit to Fly	State of Registry or State of Design	Allows non-certified prototype aircraft or aircraft with minor airworthiness defects to fly for flight testing, prototype evaluation, or ferry flight to a maintenance facility.
2. Type Certificate (TC)	State of Design / Manufacture	Certifies that the design (model) of the aircraft meets all applicable Airworthiness Codes (e.g. EASA CS-25 for large aeroplanes, CS-23 for light aeroplanes). Held by the manufacturer (Airbus, Boeing).
3. Supplemental TC (STC)	State of Design	Issued to an organization that introduces major modifications or alterations to an existing certified aircraft type (e.g. winglet installation, cargo conversion, glass cockpit upgrade).
4. Certificate of Airworthiness (C of A)	State of Registry	Issued to EACH INDIVIDUAL aircraft certifying it conforms to the approved Type Certificate and is in condition for safe operation. Must be carried on board on every international flight (Art. 29).

[!] EXAM TRAP (AVIATIONEXAM) - State of Manufacture vs State of Registry

AviationExam Trap: Who issues what? - The TYPE CERTIFICATE (TC) is issued by the STATE OF DESIGN / MANUFACTURE (e.g. France for Airbus, USA for Boeing). - The CERTIFICATE OF AIRWORTHINESS (C of A) for each individual aircraft is issued or validated by the STATE OF REGISTRY.

3. CONTINUING AIRWORTHINESS: EASA PART-M, PART-CAMO & PART-145

A Certificate of Airworthiness remains valid only if continuing airworthiness is maintained in accordance with EASA Part-M:

- Airworthiness Directives (AD): Legally enforceable rules issued by the State of Design or EASA to correct an unsafe condition found in an aircraft type. ADs are MANDATORY. Failure to comply with an AD invalidates the C of A immediately, grounding the aircraft.
- Service Bulletins (SB): Technical documents issued by aircraft or engine manufacturers suggesting improvements, modifications, or specialized inspections. SBs are NON-MANDATORY by default, unless mandated by an Airworthiness Directive.
- Certificate of Release to Service (CRS): Issued by appropriately licensed Part-145 certifying staff after any maintenance, overhaul, repair, or modification before the aircraft is permitted to fly.
- Airworthiness Review Certificate (ARC - EASA Form 15a/b/c): Under EASA Part-M, a standard C of A has an unlimited duration, but is legally valid ONLY when accompanied by a valid ARC, which must be renewed ANNUALLY (1-year validity). Form 15a is issued by the Authority; Form 15b is issued by an approved CAMO; Form 15c by a CAO.

4. MINIMUM EQUIPMENT LIST (MEL) & CDL ARCHITECTURE

Commercial transport aircraft may depart with inoperative instruments or equipment ONLY under strict approved relief rules:

- MMEL (Master Minimum Equipment List): Established by the manufacturer and approved by the State of Design. Lists items that may temporarily be inoperative under specified operating conditions.
- MEL (Minimum Equipment List): Established by the OPERATOR for each aircraft type, based on the MMEL, and approved by the State of the Operator. The MEL can be MORE restrictive than the MMEL, but NEVER LESS restrictive.
- Configuration Deviation List (CDL): Lists external missing aerodynamic or structural parts (e.g. flap track fairings, landing gear doors, static wicks) permitted for dispatch, along with associated aircraft performance penalties.

MEL Rectification Interval Categories (AviationExam Core)

Rectification Category	Mandatory Legal Repair Time Limit
Category A	No standard calendar time limit. Must be repaired within the specific time interval listed in the remarks column of the MEL (e.g. 5 flight cycles, 10 flight hours).
Category B	Must be rectified within THREE (3) consecutive calendar days (72 hours), excluding the day the defect was entered in the logbook.
Category C	Must be rectified within TEN (10) consecutive calendar days (240 hours), excluding the day the defect was discovered.
Category D	Must be rectified within ONE HUNDRED AND TWENTY (120) consecutive calendar days, excluding the day of discovery.

5. MAINTENANCE RECORDS & COMPONENT RELEASE (EASA FORM 1)

Continuing airworthiness records must be preserved to prove compliance: - Maintenance records must be retained until 36 MONTHS after the aircraft has been permanently withdrawn from service. - EASA Form 1: The official Authorized Release Certificate certifying that a component or part has been manufactured or maintained in accordance with approved data.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA Part-M, what is the validity period of an Airworthiness Review Certificate (ARC)?

[A] 6 months [B] 1 year [C] 2 years [D] Unlimited provided maintenance is carried out

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The ARC is valid for 1 year and must be re-issued annually by an approved CAMO or authority.

Q2: What is the legal status of an Airworthiness Directive (AD) versus a Service Bulletin (SB)?

[A] Both are voluntary recommendations. [B] SBs are mandatory; ADs are optional. [C] ADs are mandatory; SBs are non-mandatory unless mandated by an AD. [D] ADs only apply to military aircraft.

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. An Airworthiness Directive is legally binding and mandatory. A Service Bulletin is recommended by the manufacturer but only becomes legally mandatory if incorporated into an AD.

Q3: Can an operator depart with an item inoperative that is NOT listed in the MEL?

[A] Yes, at the discretion of the Commander. [B] Yes, if operating in daylight VMC. [C] NO, all inoperative items must be covered by relief in the MEL or repaired before flight. [D] Yes, for domestic ferry flights.

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. Under Part-CAT, an aircraft cannot be dispatched with an inoperative item unless specifically permitted by the approved MEL.

Q4: What is the maximum allowable rectification interval for Category C MEL items?

[A] 3 calendar days (72 hours) [B] 10 calendar days (240 hours) [C] 30 calendar days [D] 120 calendar days

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Category C items must be rectified within 10 consecutive calendar days excluding the day of discovery.

Q5: What document certifies that an individual aircraft component has been maintained to approved data?

[A] EASA Form 1 [B] EASA Form 15b [C] Certificate of Airworthiness [D] Journey Log Book

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [A]. EASA Form 1 is the official Authorised Release Certificate for aircraft components and parts.